Quadratics: forms and methods

Lecture and worked examples

Chapter path

Objectives. Choose a form suited to a question; switch forms in simple cases; prove and use the root formula; justify signs and solve inequalities; optimise while respecting the domain.

Prerequisites. Expansion, factorisation, special products, zero-product property, signs of products, square roots and reading graphs.

Before quadratics

Expand $(x-3)^2$. Factorise x^2-16. Solve $(x-2)(x+5)=0$, then $x^2=9$ over $\mathbb{R}$. Give the sign of $2x-6$. Find the minimum of $(x-1)^2+4$.

Worked answer: Before quadratics

$(x-3)^2 = x^2 - 6x + 9$. $x^2 - 16 = (x-4)(x+4)$. The zero-product property gives $x = 2$ or $x = -5$. A real number with square 9 is -3 or 3. The sign of $2(x-3)$ is negative before 3, zero at 3 and positive after 3. The minimum is 4, attained only at 1, because a square is nonnegative.

Explore a rectangle

Build a rectangle of perimeter 24 units. Choose several positive widths that give a rectangle, including 2, 4, 6, 8 and 10. Calculate the matching length and area. Which width seems to give the largest area? State a conjecture, then seek a justification valid for every width.

Worked answer: Explore a rectangle

The length is $12 - x$, with $0 < x < 12$. Thus $A(x) = 36 - (x-6)^2 \leq 36$, with equality at $x = 6$. The length is also 6, so the maximum-area rectangle is a square. Experiments suggest the result; the completed square proves it for every admissible width.

Three forms, three readings

A quadratic polynomial has the form $f(x) = ax^2 + bx + c$, with $a \neq 0$. If $a = 0$, the equation changes type and this chapter's formulas do not apply.

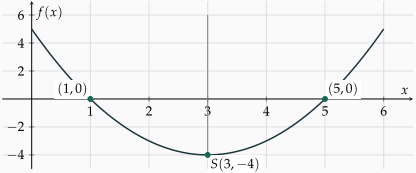
Form	What it reveals
$ax^2 + bx + c$	The value at zero is c.
$a(x - \alpha)^2 + \beta$	The vertex is (α, β); β is the extremum on $\mathbb{R}$.
$a(x - r_1)(x - r_2)$	The roots and sign, when real factorisation is possible.

These expressions describe the same function. Choose the one that makes the question easiest to answer.

Three forms, three readings

For $f(x) = x^2 - 6x + 5$, complete the square: $x^2 - 6x = (x-3)^2 - 9$. Thus $f(x) = (x-3)^2 - 4$. A square is nonnegative, so $f(x) \geq -4$, with equality only at $x = 3$.

For the zeros, use a difference of squares: $(x-3)^2 - 2^2 = (x-5)(x-1)$. The product vanishes exactly at 1 or 5. The expanded form gives $f(0) = 5$. Check every transformation by expanding.



Three forms, three readings

A factorised expression is not obtained by putting each term separately in brackets.

Three forms, three readings

Give three forms of $g(x) = x^2 + 2x - 3$. Read off its roots and its minimum.

Worked answer: Three forms, three readings

1. Complete the square: $x^2 + 2x - 3 = (x+1)^2 - 4$.
2. Factorise the difference of squares to obtain $(x-1)(x+3)$.
3. The roots are -3 and 1; the minimum is -4, attained at -1.
4. Expanding either form gives $x^2 + 2x - 3$.

Discriminant and solving

For $ax^2 + bx + c = 0$ with $a \neq 0$, let $\Delta = b^2 - 4ac$. If $\Delta < 0$, there is no real root; if $\Delta = 0$, there is one root $-\dfrac{b}{2a}$; if $\Delta > 0$, there are two roots $\dfrac{-b - \sqrt{\Delta}}{2a}$ and $\dfrac{-b + \sqrt{\Delta}}{2a}$. To order the roots, take the sign of a into account.

Discriminant and solving

Proof. For $a \neq 0$, multiply the equation by $4a$. This is reversible and gives $4a^2x^2 + 4abx + 4ac = 0$, hence $(2ax + b)^2 = b^2 - 4ac = \Delta$.

- If $\Delta < 0$, no real square works.
- If $\Delta = 0$, $2ax + b = 0$, giving $x = -\dfrac{b}{2a}$.
- If $\Delta > 0$, $2ax + b = \pm\sqrt{\Delta}$; solving for x gives the two usual roots.

Every step is an equivalence, so all solutions have been found. Order the two roots before building a sign table.

Discriminant and solving

$2x^2 + x - 3 = 0$: $\Delta = 1 + 24 = 25$; $x = \dfrac{-1 \pm 5}{4}$, that is 1 and $-\dfrac{3}{2}$. Check: $(x-1)(2x+3) = 2x^2 + x - 3$.

Discriminant and solving

The denominator is $2a$, not 2; the sign of b is kept with its brackets in b^2.

Discriminant and solving

Solve $3x^2 - 6x + 3 = 0$ and $x^2 + 2x + 5 = 0$.

Worked answer: Discriminant and solving

$3x^2 - 6x + 3 = 3(x-1)^2$, so the only root is 1; its discriminant is 0. For $x^2 + 2x + 5$, the discriminant is -16, so there is no real root. Independently, $(x+1)^2 + 4 \geq 4$, ruling out a zero.

Factorising without automating the discriminant

If two real roots exist, their sum is $-\frac{b}{a}$ and their product is $\frac{c}{a}$. An obvious root or a special product often allows a shorter factorisation. The polynomials of degree 2 that vanish at two distinct reals r and s are exactly $a(x-r)(x-s)$ with $a \neq 0$.

Factorising without automating the discriminant

Let $P(x) = x^2 - 7x + 12$. The numbers 3 and 4 have sum 7 and product 12, hence $P(x) = (x-3)(x-4)$. **A separate question:** find f with roots 3 and 4 and $f(0) = 24$. Write $f(x) = a(x-3)(x-4)$. Then $12a = 24$, giving $f(x) = 2(x-3)(x-4)$. This f is a different function from P.

Factorising without automating the discriminant

The sum of the roots is $-\frac{b}{a}$; do not forget the minus sign. A single root is not enough to determine a.

Factorising without automating the discriminant

Determine the polynomial f of degree 2 such that $f(1) = f(4) = 0$ and $f(0) = -8$.

Worked answer: Factorising without automating the discriminant

The distinct roots 1 and 4 give $f(x) = a(x-1)(x-4)$. Then $f(0) = 4a = -8$, giving $a = -2$. Thus $f(x) = -2(x-1)(x-4) = -2x^2 + 10x - 8$. Substitution verifies all three conditions.

Vertex, monotonicity and domain

In $a(x-\alpha)^2+\beta$, the square measures how far x lies from α. It decreases as x approaches α, then increases.

For $a>0$, f decreases on $(-\infty,\alpha]$ and increases on $[\alpha,+\infty)$; the minimum is β. For $a<0$, multiplying by a reverses the order, giving increasing then decreasing behaviour and a maximum β.

Vertex, monotonicity and domain (2)

On a restricted interval, first check whether α is allowed. Otherwise use monotonicity to compare admissible values. An extremum on $\mathbb{R}$ need not be attained on the problem's domain.

For two distinct roots $r < s$, the quadratic has the sign of a outside $[r, s]$ and the opposite sign between the roots. It is zero at the roots. With no real root, it keeps the sign of a. With a double root, it keeps the sign of a away from that root. Optimisation is done on the domain of the problem, which may limit access to the vertex.

Sign, inequalities and optimisation

For $r < s$, the factors $x - r$ and $x - s$ are both negative before r, opposite in sign between the roots, and both positive after s. Their product is positive outside and negative inside; multiplying by a gives the quadratic's sign.

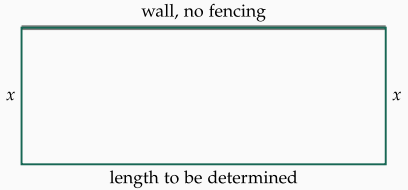
Rebuild this argument from the factors rather than memorising a rule. Strict inequalities exclude roots; non-strict inequalities may include them. Intersect the solution set with the problem's domain.

$-2(x-1)(x-5) \geq 0$ is equivalent to $x \in [1,5]$. For $A(x) = x(12-x)$ on $(0,12)$, $A(x) = 36 - (x-6)^2 \leq 36$, with equality at 6. A rectangle with perimeter 24 m therefore has a maximum area of 36 m^2 when it is a square.

In a strict inequality, the roots are excluded. Check the geometric bounds before concluding.

Sign, inequalities and optimisation

Solve $x^2 - 4x - 5 < 0$, then maximise $x(8 - x)$ on $[0, 3]$.



Worked answer: Sign, inequalities and optimisation

$x^2 - 4x - 5 = (x+1)(x-5)$. Between -1 and 5 the factors have opposite signs, so the strict inequality holds on $(-1, 5)$, excluding endpoints. For $A(x) = 16 - (x-4)^2$, the vertex 4 is outside $[0, 3]$. The distance to 4 decreases up to 3, so the maximum is $A(3) = 15$, not 16.

Review: choose and justify

1. Solve $x^2 - 9 = 0$ with a short method. 2. For $h(x) = -2(x-1)^2 + 8$, find the maximum on $\mathbb{R}$ and its roots. 3. Can $(x-1)^2 < 0$ have a real solution? Justify.

Worked answer: Review: choose and justify

1. Factor to get roots -3 and 3. 2. Since a square is nonnegative, $h(x) \leq 8$, attained at 1. Its zeros satisfy $(x-1)^2 = 4$, giving -1 and 3. 3. There is no real solution because a square cannot be negative.